

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Bachelor of Technology (Computer Science and Engineering) SEMESTER - 3 Winter 2025 (Regular)
Course : Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology
Semester : SEMESTER - 3
Subject Code & Name: 25AF12051PC302 - DATA STRUCTURES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is an Abstract Data Type?
a) Array ~~b) Stack~~ c) Integer ~~d) Pointer~~
- 2 Sparse matrices are mainly used to:
~~a) Reduce storage space and time complexity~~ b) Increase redundancy in storage
c) Store only square matrices d) Speed up sequential access
- 3 The postfix expression of $(A + B) * C$ is ____
~~a) AB+C*~~ b) ABC+* c) A+BC* d) AB*C+
- 4 In a circular queue, the condition for overflow is ____
a) front = rear ~~b) rear = front + 1 (mod size)~~ c) front = -1
d) rear = -1
- 5 Time complexity of binary search is:
a) $O(n)$ ~~b) $O(\log n)$~~ c) $O(n^2)$ d) $O(1)$
- 6 Which data structure follows FIFO principle?
a) Stack ~~b) Queue~~ c) Tree d) Graph
- 7 Collision in hash table occurs when
a) Table is full b) Keys are same ~~c) Hash function gives same~~
address d) Load factor is zero
- 8 In linked list, insertion is efficient because
~~a) No shifting~~ b) Random access c) Fixed size d) Indexing
- 9 Garbage collection is related to
a) Stack b) Queue ~~c) Linked list~~ d) Tree
- 10 In Binary Search Tree, inorder traversal gives
a) Random order b) Reverse order ~~c) Sorted order~~ d) Level order
- 11 Graph representation using adjacency matrix requires
a) $O(V)$ b) $O(E)$ ~~c) $O(V^2)$~~ d) $O(\log V)$
- 12 Best case time complexity of insertion sort is
a) $O(n^2)$ b) $O(\log n)$ ~~c) $O(n)$~~ d) $O(1)$

Q2. Solve the following.

- A) Consider an array named B [0:15, 0:5]. Assume that base address is 1003 and 4 bytes reserved for one element. Compute the following: 6
- i) The number of elements in array B.
 - ii) The memory space occupied by array B
 - iii) The memory address of B[7, 2], assume column major storage of an array in memory.
- B) Define the following terms: 6
- 1) Abstract Data Types 2) Hash Function 3) Sparse Matrix

Q3. Solve the following.

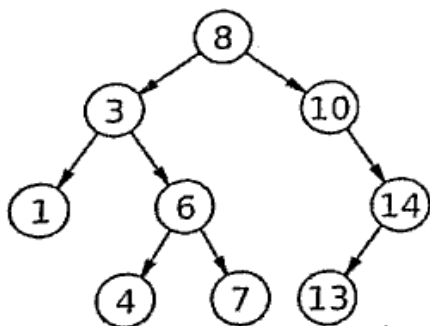
- A) Convert the expression $((A + B) * C - (D - E) ^ (F + G))$ to equivalent Prefix and Postfix notations. 6
- B) Explain stack with example. Write algorithm for Push() and Pop() operation. 6

Q4. Solve Any Two of the following.

- A) What is dynamic memory management? Explain garbage collection. 6
- B) Write an algorithm for performing following operations in the Singly Linked List: 6
- a) Insert the elements at the end
 - b) Delete the element from the beginning
- C) Explain circular linked list in details with advantages. 6

Q5. Solve Any Two of the following.

- A) Explain Warshall's algorithm with example. 6
- B) Explain Binary Search Tree. For following binary tree, give preorder, inorder and postorder traversal sequence. 6



- C) Following numbers are inserted into empty binary search tree. Draw the final Binary Search tree T. 6
- 25, 20, 10, 36, 22, 5, 1, 8, 30, 12, 15, 40, 28, 38, 48, 45, 50

Q6. Solve Any Two of the following.

- A) Write an algorithm for Linear Search and Binary Search 6
- B) What is skip list? Explain its operations. 6
- C) Sort the following list using radix sort. Show all the passes neatly: 3, 45, 7, 18, 9, 4, 89, 103, 11, 21

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